

E-Commerce Sales Dashboard & Analytics

End-to-End Project | PostgreSQL • Python • Power BI

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30,000 Records

Transaction-level data spanning four years, normalized and cataloged.



41 Features

Detailed feature set: product, customer, order, shipping, marketing, and financial fields.



5 Dashboard Pages

Executive to operational views — actionable, interactive, and deployable.

Business Problem Statement

Four core challenges limiting revenue visibility, margin control, and decision speed.

Revenue Leakage

Unable to isolate categories or countries driving net gains vs. losses; limited attribution across channels.

High Return Rate — 10.11%

~3,033 returned orders with no structured root-cause analysis across SKUs, sizes, or fulfillment centers.

Discount Strategy Unclear

Coupons average 12.62% discount; unclear impact on profitability and customer lifetime value.

Siloed Data

30K records remain in raw CSVs — no centralized, governed dashboard for stakeholders.



These problems motivated a structured analytics pipeline: ingestion → modeling → visualization → action.

Project Objectives & Approach

Objectives

- Analyze 30K transactions to surface revenue drivers
- Identify root causes for 10.11% returns
- Quantify coupon impact on margins
- Segment customers & prioritize retention
- Deliver a 5-page interactive Power BI dashboard

Methodology



Pipeline: ingest raw CSVs, model in PostgreSQL, run analytic queries, explore with Python, publish in Power BI.

Tools & Technologies

Core stack that powered the analysis and delivery.

1

PostgreSQL + pgAdmin4

Database schema design, ETL ingestion, 15+ business queries, CTEs and Window functions for time-series and cohort analysis.



2

Python / Jupyter

Pandas & NumPy for data prep, Seaborn & Matplotlib for visualization – 17-step EDA yielding 21 charts for hypothesis testing.



3

Power BI Desktop

Live DB connection, 5 interactive pages, 15+ DAX measures for KPIs, slicers, and drill-through experiences.



Key Performance Indicators

Concise KPI grid for executive monitoring – value, context, and trend callouts.



\$11.37M

Total Revenue (all 4 years)



30K

Total Orders



9K

Unique Customers – Avg 3.3 orders/customer



10.11%

Return Rate – 3,033 orders



\$2.22M

Total Profit – 19.49% Margin

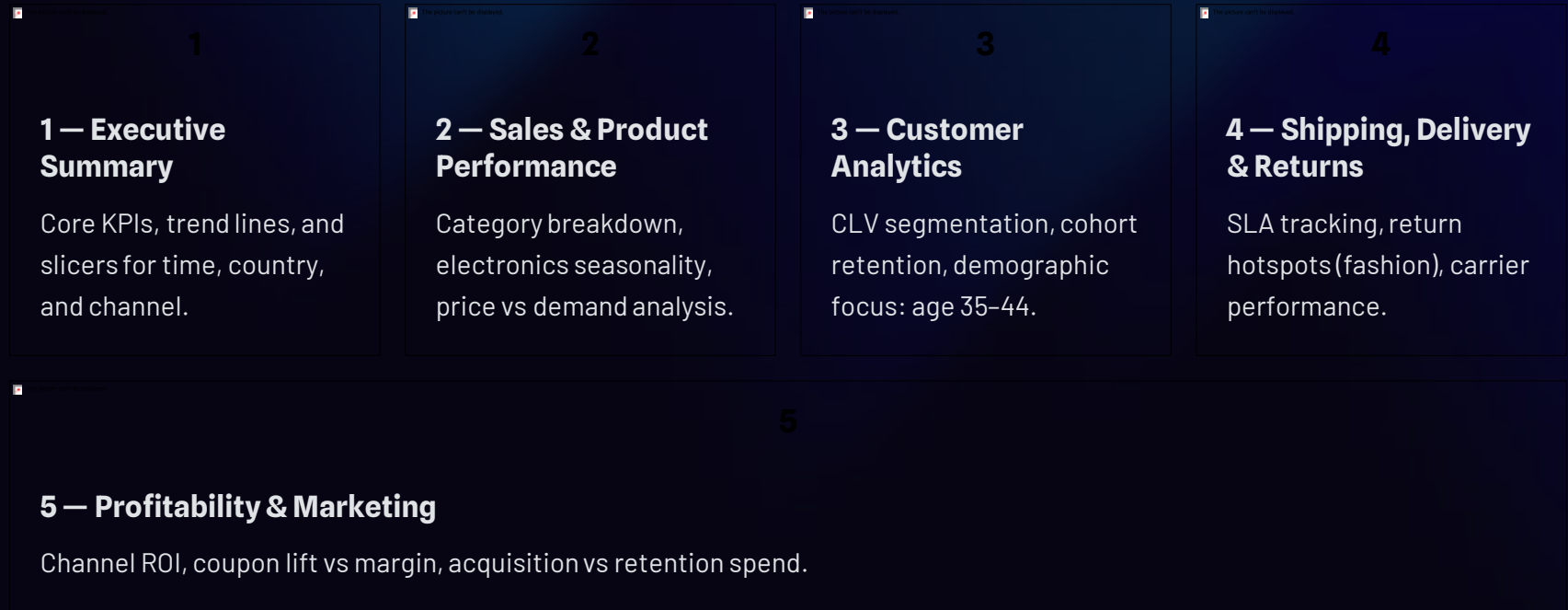


4.44

Avg Delivery Days

Power BI Dashboard Architecture

Five-purpose pages – designed for decision cadence from executive to operations.



SQL Analysis Highlights

Six core analytic queries implemented in PostgreSQL to operationalize insights.

Q1 — Business Overview

Aggregate revenue, profit, orders, returns by month & country.

Q2 — Top Profitable Categories

Category-level margin breakdown using grouped aggregations and joins.

Q3 — MoM Growth via LAG()

Month-over-month trends using window functions and percent change calculations.

Q4 — Return Rate via CASE WHEN

Return flags and rate calculations segmented by product and fulfillment center.

Q5 — Customer Ranking via RANK()

RFM-style ranking and top-customer identification with windowed ranking.

Q6 — Discount Impact Analysis

Coupon vs non-coupon cohorts; pivoted profitability and AOV comparisons.

All queries are parameterized for fast dashboard refresh and scheduled materialized views.

Key Business Insights

Insight — Revenue Concentration

Electronics contributes ~55% of revenue (~\$6.3M). Prioritize inventory and pricing strategies there.

Insight — Return Hotspot

Fashion shows the highest product return rates; size-fit and variant issues are primary drivers.

Insight — Channel Performance

Social Media is the top acquisition channel (~\$2M revenue); outperforms paid search on ROI.

Insight — Discount Effects

Blanket coupons reduce AOV and compress margins; targeted coupons show better retention lift.

Recommendation (Rec): Implement targeted coupon experiments, tighten fashion returns controls, and allocate marketing budget by channel ROI.

Strategic Business Recommendations

Prioritized actions with urgency tags to guide execution over the next 90–180 days.

High Priority

- Focus Electronics campaigns in UK & Saudi Arabia to capitalize on demand pockets.
- Revamp Fashion size guides and introduce virtual fit tools to reduce returns.

Medium Priority

- Restrict coupons to Gold/Platinum tiers and A/B test targeted promotions.
- Reallocate incremental spend to Social Media channels with proven ROI.

Low Priority

- Optimize Same-Day delivery SLAs in high-value regions to improve NPS and repeat purchases.



Action owners: Marketing, Merchandising, Fulfillment. Track progress via dashboard weekly.